10th grade
Learning evidence 2.1 Decision analysis supplementary solutions 1

Problems

Problem 1: ## Problem description

A weekend coffee cart must choose a service plan for a street market. Plan A is a specialty menu and Plan B is a classic menu. Customer turnout can be high or low with probabilities $0{,}55$ and $0{,}45$. Profit outcomes are given directly: under high turnout Plan A earns 700 and Plan B earns 480, and under low turnout Plan A earns 250 and Plan B earns 200. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Plan A (specialty menu), Plan B (classic menu)
States of Nature	Turnout high or low with probabilities $0{,}55$ and $0{,}45$
Events	Realized customer turnout in the market day
Consequences	Profit in dollars

C3

Payoff is the profit in dollars given for each alternative and state.

State of nature	Probability	Plan A	Plan B
High turnout	0.55	700	480
Low turnout	0.45	250	200

C4

$$\text{máx}(\text{Plan A}) = \text{máx}\{700, 250\} = 700,$$
$$\text{máx}(\text{Plan B}) = \text{máx}\{480, 200\} = 480.$$

$$\text{máx}\{\text{máx}(\text{Plan A}), \text{máx}(\text{Plan B})\} = \text{máx}\{700, 480\} = 700.$$

The Maximax choice is Plan A with 700. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\text{mín(Plan A)} = \text{mín}\{700, 250\} = 250,$$
$$\text{mín(Plan B)} = \text{mín}\{480, 200\} = 200.$$
$$\text{máx}\{\text{mín(Plan A)}, \text{mín(Plan B)}\} = \text{máx}\{250, 200\} = 250.$$

The Maximin choice is Plan A with 250 because it has the least severe worst case. This is *safe* because it protects against the worst outcome.

C1

$$EV_A = 0{,}55(700) + 0{,}45(250) = 385 + 112{,}5 = 497{,}5,$$
$$EV_B = 0{,}55(480) + 0{,}45(200) = 264 + 90 = 354.$$

The expected value criterion chooses Plan A because $497{,}5$ is the largest expected profit. This aligns with the Maximax and Maximin outcomes for Plan A, so it is the balanced decision using the probabilities.

Problem 2: Problem description

A campus bookstore must choose a stocking plan for a semester. Plan A is premium notebooks, Plan B is standard notebooks, and Plan C is discount notebooks. Demand for notebooks can be strong, moderate, or weak with probabilities $0{,}35$, $0{,}45$, and $0{,}20$. Expected unit sales depend on demand and plan. Under strong demand Plan A sells 420 units, Plan B sells 520 units, and Plan C sells 600 units. Under moderate demand Plan A sells 300 units, Plan B sells 380 units, and Plan C sells 450 units. Under weak demand Plan A sells 180 units, Plan B sells 240 units, and Plan C sells 300 units. Unit prices are 22, 18, and 15 for Plans A, B, and C. Use these parameters to calculate the payoff table. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Plan A (premium), Plan B (standard), Plan C (discount)
States of Nature	Demand strong, moderate, weak with probabilities $0{,}35$, $0{,}45$, $0{,}20$
Events	Realized demand level during the semester
Consequences	Payoff in dollars from revenue

C3

Payoff equals revenue, where revenue = units $\times$ price.

Units by plan and demand

Plan	Strong	Moderate	Weak
Plan A	420	300	180
Plan B	520	380	240
Plan C	600	450	300

Price parameters by plan

Plan	Price per unit
Plan A	22
Plan B	18
Plan C	15

Revenue by demand

State of nature	Plan A revenue	Plan B revenue	Plan C revenue
Strong demand	$420 \cdot 22$ $= 9240$	$520 \cdot 18$ $= 9360$	$600 \cdot 15$ $= 9000$
Moderate demand	$300 \cdot 22$ $= 6600$	$380 \cdot 18$ $= 6840$	$450 \cdot 15$ $= 6750$
Weak demand	$180 \cdot 22$ $= 3960$	$240 \cdot 18$ $= 4320$	$300 \cdot 15$ $= 4500$

Payoff equals revenue, so the payoff table matches the revenue values.

Payoff table

State of nature	Probability	Plan A	Plan B	Plan C
Strong demand	0.35	9240	9360	9000
Moderate demand	0.45	6600	6840	6750
Weak demand	0.20	3960	4320	4500

C4

$$\text{máx}(\text{Plan A}) = \text{máx}\{9240, 6600, 3960\} = 9240,$$
$$\text{máx}(\text{Plan B}) = \text{máx}\{9360, 6840, 4320\} = 9360,$$
$$\text{máx}(\text{Plan C}) = \text{máx}\{9000, 6750, 4500\} = 9000.$$
$$\text{máx}\{\text{máx}(\text{Plan A}), \text{máx}(\text{Plan B}), \text{máx}(\text{Plan C})\} = \text{máx}\{9240, 9360, 9000\} = 9360.$$

The Maximax choice is Plan B with 9360. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\text{mín}(\text{Plan A}) = \text{mín}\{9240, 6600, 3960\} = 3960,$$
$$\text{mín}(\text{Plan B}) = \text{mín}\{9360, 6840, 4320\} = 4320,$$
$$\text{mín}(\text{Plan C}) = \text{mín}\{9000, 6750, 4500\} = 4500.$$
$$\text{máx}\{\text{mín}(\text{Plan A}), \text{mín}(\text{Plan B}), \text{mín}(\text{Plan C})\} = \text{máx}\{3960, 4320, 4500\} = 4500.$$

The Maximin choice is Plan C with 4500 because it gives the best worst case outcome. This is *safe* because it prioritizes the least harmful result.

C1

$$EV_A = 0{,}35(9240) + 0{,}45(6600) + 0{,}20(3960) = 3234 + 2970 + 792 = 6996,$$
$$EV_B = 0{,}35(9360) + 0{,}45(6840) + 0{,}20(4320) = 3276 + 3078 + 864 = 7218,$$
$$EV_C = 0{,}35(9000) + 0{,}45(6750) + 0{,}20(4500) = 3150 + 3037{,}5 + 900 = 7087{,}5.$$

The expected value criterion chooses Plan B because 7218 is the largest expected payoff. This matches the Maximax outcome and differs from the Maximin outcome, so it is the balanced decision using the probabilities.

Problem 3: Problem description

A meal subscription service must choose a weekly menu plan for the next quarter. Plan A is a chef series, Plan B is a family bundle, and Plan C is a value bundle. Demand can be surge, high, moderate, or low with probabilities $0{,}20$, $0{,}35$, $0{,}30$, and $0{,}15$. Expected subscriptions depend on demand and plan. Under surge demand Plan A sells 500 subscriptions, Plan B sells 620 subscriptions, and Plan C sells 700 subscriptions. Under high demand Plan A sells 420 subscriptions, Plan B sells 520 subscriptions, and Plan C sells 600 subscriptions. Under moderate demand Plan A sells 300 subscriptions, Plan B sells 380 subscriptions, and Plan C sells 450 subscriptions. Under low demand Plan A sells 200 subscriptions, Plan B sells 260 subscriptions, and Plan C sells 320 subscriptions. Unit prices are 24, 19, and 16 for Plans A, B, and C. Variable costs per subscription are 11, 9, and 7. Use these parameters to calculate the profit payoff table. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Plan A (chef series), Plan B (family bundle), Plan C (value bundle)
States of Nature	Demand surge, high, moderate, low with probabilities $0{,}20$, $0{,}35$, $0{,}30$, $0{,}15$
Events	Realized demand level for the quarter
Consequences	Profit in dollars after revenue and costs

C3

Payoff = revenue $-$ cost, where revenue = (subscriptions $\times$ price) and cost = (subscriptions $\times$ variable cost).

Subscriptions by plan and demand

Plan	Surge	High	Moderate	Low
Plan A	500	420	300	200
Plan B	620	520	380	260
Plan C	700	600	450	320

Price parameters by plan

Plan	Price per subscription
Plan A	24
Plan B	19
Plan C	16

Cost parameters by plan

Plan	Variable cost
Plan A	11
Plan B	9
Plan C	7

Revenue by demand

State of nature	Plan A revenue	Plan B revenue	Plan C revenue
Surge demand	$500 \cdot 24$ $= 12000$	$620 \cdot 19$ $= 11780$	$700 \cdot 16$ $= 11200$
High demand	$420 \cdot 24$ $= 10080$	$520 \cdot 19$ $= 9880$	$600 \cdot 16$ $= 9600$
Moderate demand	$300 \cdot 24$ $= 7200$	$380 \cdot 19$ $= 7220$	$450 \cdot 16$ $= 7200$
Low demand	$200 \cdot 24$ $= 4800$	$260 \cdot 19$ $= 4940$	$320 \cdot 16$ $= 5120$

Cost by demand

State of nature	Plan A cost	Plan B cost	Plan C cost
Surge demand	$500 \cdot 11$ $= 5500$	$620 \cdot 9$ $= 5580$	$700 \cdot 7$ $= 4900$
High demand	$420 \cdot 11$ $= 4620$	$520 \cdot 9$ $= 4680$	$600 \cdot 7$ $= 4200$
Moderate demand	$300 \cdot 11$ $= 3300$	$380 \cdot 9$ $= 3420$	$450 \cdot 7$ $= 3150$
Low demand	$200 \cdot 11$ $= 2200$	$260 \cdot 9$ $= 2340$	$320 \cdot 7$ $= 2240$

Profit parameters by plan

State of nature	Probability	Plan A	Plan B	Plan C
Surge demand	0,20	$12000 - 5500$ $= 6500$	$11780 - 5580$ $= 6200$	$11200 - 4900$ $= 6300$
High demand	0,35	$10080 - 4620$ $= 5460$	$9880 - 4680$ $= 5200$	$9600 - 4200$ $= 5400$
Moderate demand	0,30	$7200 - 3300$ $= 3900$	$7220 - 3420$ $= 3800$	$7200 - 3150$ $= 4050$
Low demand	0,15	$4800 - 2200$ $= 2600$	$4940 - 2340$ $= 2600$	$5120 - 2240$ $= 2880$

Payoff table

State of nature	Probability	Plan A	Plan B	Plan C
Surge demand	0.20	6500	6200	6300
High demand	0.35	5460	5200	5400
Moderate demand	0.30	3900	3800	4050
Low demand	0.15	2600	2600	2880

C4

$$\begin{aligned}
\text{máx}(\text{Plan A}) &= \text{máx}\{6500, 5460, 3900, 2600\} = 6500,\\
\text{máx}(\text{Plan B}) &= \text{máx}\{6200, 5200, 3800, 2600\} = 6200,\\
\text{máx}(\text{Plan C}) &= \text{máx}\{6300, 5400, 4050, 2880\} = 6300.
\end{aligned}$$

$$\text{máx}\{\text{máx}(\text{Plan A}), \text{máx}(\text{Plan B}), \text{máx}(\text{Plan C})\} = \text{máx}\{6500, 6200, 6300\} = 6500.$$

The Maximax choice is Plan A with 6500. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\begin{aligned}
\text{mín}(\text{Plan A}) &= \text{mín}\{6500, 5460, 3900, 2600\} = 2600,\\
\text{mín}(\text{Plan B}) &= \text{mín}\{6200, 5200, 3800, 2600\} = 2600,\\
\text{mín}(\text{Plan C}) &= \text{mín}\{6300, 5400, 4050, 2880\} = 2880.
\end{aligned}$$

$$\text{máx}\{\text{mín}(\text{Plan A}), \text{mín}(\text{Plan B}), \text{mín}(\text{Plan C})\} = \text{máx}\{2600, 2600, 2880\} = 2880.$$

The Maximin choice is Plan C with 2880 because it has the least severe worst case. This is *safe* because it protects against the worst outcome.

C1

Expected values use the demand probabilities.

$$\begin{aligned}
EV_A &= 0{,}20(6500) + 0{,}35(5460) + 0{,}30(3900) + 0{,}15(2600)\\
&= 1300 + 1911 + 1170 + 390 = 4771,\\
EV_B &= 0{,}20(6200) + 0{,}35(5200) + 0{,}30(3800) + 0{,}15(2600)\\
&= 1240 + 1820 + 1140 + 390 = 4590,\\
EV_C &= 0{,}20(6300) + 0{,}35(5400) + 0{,}30(4050) + 0{,}15(2880)\\
&= 1260 + 1890 + 1215 + 432 = 4797.
\end{aligned}$$

The expected value criterion chooses Plan C because 4797 is the largest expected profit. This differs from the Maximax choice and matches the Maximin choice, so the expected value emphasizes weighted outcomes.

Problem 4: ## Problem description

A drone delivery service must choose a routing plan for the next season. Strategy A uses an in house routing system and Strategy B uses a contracted routing partner. Demand can be strong or weak with probabilities $0{,}55$ and $0{,}45$. Battery cost conditions can be favorable or unfavorable with probabilities $0{,}70$ and $0{,}30$, and these conditions are independent of demand. Revenue in thousands of dollars depends on demand and strategy. Under strong demand revenue is 620 for Strategy A and 660 for Strategy B. Under weak demand revenue is 340 for Strategy A and 320 for Strategy B. Operating costs in thousands of dollars depend on battery cost conditions and strategy. Under favorable costs operating costs are 260 for Strategy A and 300 for Strategy B. Under unfavorable costs operating costs are 340 for Strategy A and 380 for Strategy B. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Strategy A (in house), Strategy B (contracted)
States of Nature	Demand strong or weak with battery costs favorable or unfavorable
Events	Realized demand level paired with battery cost conditions
Consequences	Profit in thousands of dollars from revenue minus operating costs

C3

Payoff equals revenue minus cost, where revenue is determined by demand and cost is determined by battery conditions.

Revenue parameters by strategy

Strategy	Strong	Weak
Probabilities	0.55	0.45
Strategy A	620	340
Strategy B	660	320

Cost parameters by strategy

Strategy	Favorable	Unfavorable
Probabilities	0.70	0.30
Strategy A	260	340
Strategy B	300	380

Profit is computed as revenue minus cost for each alternative and state.

Profit parameters by strategy

State of nature	Probability	Strategy A	Strategy B
Strong demand, favorable cost	$0{,}55 \cdot 0{,}70$ $= 0{,}385$	$620 - 260$ $= 360$	$660 - 300$ $= 360$
Strong demand, unfavorable cost	$0{,}55 \cdot 0{,}30$ $= 0{,}165$	$620 - 340$ $= 280$	$660 - 380$ $= 280$
Weak demand, favorable cost	$0{,}45 \cdot 0{,}70$ $= 0{,}315$	$340 - 260$ $= 80$	$320 - 300$ $= 20$
Weak demand, unfavorable cost	$0{,}45 \cdot 0{,}30$ $= 0{,}135$	$340 - 340$ $= 0$	$320 - 380$ $= -60$

Combined probabilities are $0{,}55 \cdot 0{,}70 = 0{,}385$, $0{,}55 \cdot 0{,}30 = 0{,}165$, $0{,}45 \cdot 0{,}70 = 0{,}315$, $0{,}45 \cdot 0{,}30 = 0{,}135$.

Payoff table

State of nature	Probability	Strategy A	Strategy B
Strong demand, favorable cost	0.385	360	360
Strong demand, unfavorable cost	0.165	280	280
Weak demand, favorable cost	0.315	80	20
Weak demand, unfavorable cost	0.135	0	-60

C4

$$\text{máx}(\text{Strategy A}) = \text{máx}\{360, 280, 80, 0\} = 360,$$
$$\text{máx}(\text{Strategy B}) = \text{máx}\{360, 280, 20, -60\} = 360.$$
$$\text{máx}\{\text{máx}(\text{Strategy A}), \text{máx}(\text{Strategy B})\} = \text{máx}\{360, 360\} = 360.$$

The Maximax choice is Strategy A or Strategy B with 360. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\text{mín}(\text{Strategy A}) = \text{mín}\{360, 280, 80, 0\} = 0,$$
$$\text{mín}(\text{Strategy B}) = \text{mín}\{360, 280, 20, -60\} = -60.$$
$$\text{máx}\{\text{mín}(\text{Strategy A}), \text{mín}(\text{Strategy B})\} = \text{máx}\{0, -60\} = 0.$$

The Maximin choice is Strategy A with 0 because it has the least severe worst case. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined state probabilities.

$$\begin{aligned} EV_A &= 0{,}385(360) + 0{,}165(280) + 0{,}315(80) + 0{,}135(0) \\ &= 138{,}6 + 46{,}2 + 25{,}2 + 0 = 210, \\ EV_B &= 0{,}385(360) + 0{,}165(280) + 0{,}315(20) + 0{,}135(-60) \\ &= 138{,}6 + 46{,}2 + 6{,}3 - 8{,}1 = 183. \end{aligned}$$

The expected value criterion chooses Strategy A because 210 is the largest expected profit. This differs from the Maximin choice and matches the Maximax outcome, so the expected value gives a balanced decision using the probabilities.

Problem 5: Problem description

A regional utility must choose a generation mix for the next year. Strategy A is solar heavy, Strategy B is wind heavy, and Strategy C is balanced. Demand can be strong or weak with probabilities $0{,}55$ and $0{,}45$. Operating cost conditions can be favorable or unfavorable with probabilities $0{,}60$ and $0{,}40$, and these conditions are independent of demand. Revenue in thousands of dollars depends on demand and strategy. Under strong demand revenue is 680 for Strategy A, 620 for Strategy B, and 650 for Strategy C. Under weak demand revenue is 360 for Strategy A, 380 for Strategy B, and 370 for Strategy C. Operating costs in thousands of dollars depend on cost conditions and strategy. Under favorable costs operating costs are 300 for Strategy A, 280 for Strategy B, and 290 for Strategy C. Under unfavorable costs operating costs are 380 for Strategy A, 360 for Strategy B, and 370 for Strategy C. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Strategy A (solar), Strategy B (wind), Strategy C (balanced)
States of Nature	Demand strong or weak with cost conditions favorable or unfavorable
Events	Realized demand level paired with operating cost conditions
Consequences	Profit in thousands of dollars from revenue minus operating costs

C3

Payoff = revenue − cost, where revenue is determined by demand and cost is determined by operating conditions.

Revenue parameters by strategy

Strategy	Strong	Weak
Probabilities	0.55	0.45
Strategy A	680	360
Strategy B	620	380
Strategy C	650	370

Cost parameters by strategy

Strategy	Favorable	Unfavorable
Probabilities	0.60	0.40
Strategy A	300	380
Strategy B	280	360
Strategy C	290	370

Profit is computed as Revenue minus Cost for each alternative and state.

Profit parameters by strategy

State of nature	Probability	Strategy A	Strategy B	Strategy C
Strong demand, favorable cost	$0{,}55 \cdot 0{,}60$ $= 0{,}33$	$680 - 300$ $= 380$	$620 - 280$ $= 340$	$650 - 290$ $= 360$
Strong demand, unfavorable cost	$0{,}55 \cdot 0{,}40$ $= 0{,}22$	$680 - 380$ $= 300$	$620 - 360$ $= 260$	$650 - 370$ $= 280$
Weak demand, favorable cost	$0{,}45 \cdot 0{,}60$ $= 0{,}27$	$360 - 300$ $= 60$	$380 - 280$ $= 100$	$370 - 290$ $= 80$
Weak demand, unfavorable cost	$0{,}45 \cdot 0{,}40$ $= 0{,}18$	$360 - 380$ $= -20$	$380 - 360$ $= 20$	$370 - 370$ $= 0$

Payoff table

State of nature	Probability	Strategy A	Strategy B	Strategy C
Strong demand, favorable cost	0.33	380	340	360
Strong demand, unfavorable cost	0.22	300	260	280
Weak demand, favorable cost	0.27	60	100	80
Weak demand, unfavorable cost	0.18	-20	20	0

C4

$$\begin{aligned}\text{máx}(\text{Strategy A}) &= \text{máx}\{380, 300, 60, -20\} = 380,\\ \text{máx}(\text{Strategy B}) &= \text{máx}\{340, 260, 100, 20\} = 340,\\ \text{máx}(\text{Strategy C}) &= \text{máx}\{360, 280, 80, 0\} = 360.\end{aligned}$$

$$\text{máx}\{\text{máx}(\text{Strategy A}), \text{máx}(\text{Strategy B}), \text{máx}(\text{Strategy C})\} = \text{máx}\{380, 340, 360\} = 380.$$

The Maximax choice is Strategy A with 380. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\begin{aligned}\text{mín}(\text{Strategy A}) &= \text{mín}\{380, 300, 60, -20\} = -20,\\ \text{mín}(\text{Strategy B}) &= \text{mín}\{340, 260, 100, 20\} = 20,\\ \text{mín}(\text{Strategy C}) &= \text{mín}\{360, 280, 80, 0\} = 0.\end{aligned}$$

$$\text{máx}\{\text{mín}(\text{Strategy A}), \text{mín}(\text{Strategy B}), \text{mín}(\text{Strategy C})\} = \text{máx}\{-20, 20, 0\} = 20.$$

The Maximin choice is Strategy B with 20 because it has the least severe worst case. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined state probabilities.

$$\begin{aligned}EV_A &= 0{,}33(380) + 0{,}22(300) + 0{,}27(60) + 0{,}18(-20)\\ &= 125{,}4 + 66 + 16{,}2 - 3{,}6 = 204,\\ EV_B &= 0{,}33(340) + 0{,}22(260) + 0{,}27(100) + 0{,}18(20)\\ &= 112{,}2 + 57{,}2 + 27 + 3{,}6 = 200,\\ EV_C &= 0{,}33(360) + 0{,}22(280) + 0{,}27(80) + 0{,}18(0)\\ &= 118{,}8 + 61{,}6 + 21{,}6 + 0 = 202.\end{aligned}$$

The expected value criterion chooses Strategy A because 204 is the largest expected profit. This differs from the Maximin choice and matches the Maximax choice, so the expected value gives a balanced decision using the probabilities.

Problem 6: Problem description

A community fitness hub must choose a membership model for the next year. Model A is a weekday focus and Model B is a family pass. Demand can be high, medium, or low with probabilities $0{,}40$, $0{,}35$, and $0{,}25$. Staffing costs can be favorable or unfavorable with probabilities $0{,}55$ and $0{,}45$, and these costs are independent of demand. Expected membership counts depend on demand and model. Under high demand Model A has 700 members and Model B has 650 members. Under medium demand Model A has 520 members and Model B has 480 members. Under low demand Model A has 360 members and Model B has 320 members. Membership fees are 38 for Model A and 42 for Model B per member per year. Variable staffing cost per member is 20 when costs are favorable and 28 when costs are unfavorable. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Model A (weekday), Model B (family)
States of Nature	Demand high, medium, low with staffing costs favorable or unfavorable
Events	Realized demand level paired with staffing cost conditions
Consequences	Annual profit in dollars from membership fees minus staffing costs

C3

Payoff = revenue − cost, where revenue = (members × fee) and cost = (members × variable cost).

Members per model by demand

Model	High	Medium	Low
Model A	700	520	360
Model B	650	480	320

Cost parameters by staffing condition

Condition	Favorable	Unfavorable
Variable cost	20	28

Revenue parameters by member

Model	Fee per member
Model A	38
Model B	42

Revenue by demand

State of nature	Model A revenue	Model B revenue
High demand	$700 \cdot 38$ $= 26600$	$650 \cdot 42$ $= 27300$
Medium demand	$520 \cdot 38$ $= 19760$	$480 \cdot 42$ $= 20160$
Low demand	$360 \cdot 38$ $= 13680$	$320 \cdot 42$ $= 13440$

Cost by demand and staffing condition

State of nature	Model A cost	Model B cost
High demand, favorable cost	$700 \cdot 20$ $= 14000$	$650 \cdot 20$ $= 13000$
High demand, unfavorable cost	$700 \cdot 28$ $= 19600$	$650 \cdot 28$ $= 18200$
Medium demand, favorable cost	$520 \cdot 20$ $= 10400$	$480 \cdot 20$ $= 9600$
Medium demand, unfavorable cost	$520 \cdot 28$ $= 14560$	$480 \cdot 28$ $= 13440$
Low demand, favorable cost	$360 \cdot 20$ $= 7200$	$320 \cdot 20$ $= 6400$
Low demand, unfavorable cost	$360 \cdot 28$ $= 10080$	$320 \cdot 28$ $= 8960$

Profit is computed as Revenue minus Cost for each alternative and state.

Profit parameters by model

State of nature	Model A	Model B	Probabilities
High demand, favorable cost	$26600 - 14000$ $= 12600$	$27300 - 13000$ $= 14300$	$0{,}40 \cdot 0{,}55$ $= 0{,}22$
High demand, unfavorable cost	$26600 - 19600$ $= 7000$	$27300 - 18200$ $= 9100$	$0{,}40 \cdot 0{,}45$ $= 0{,}18$
Medium demand, favorable cost	$19760 - 10400$ $= 9360$	$20160 - 9600$ $= 10560$	$0{,}35 \cdot 0{,}55$ $= 0{,}1925$
Medium demand, unfavorable cost	$19760 - 14560$ $= 5200$	$20160 - 13440$ $= 6720$	$0{,}35 \cdot 0{,}45$ $= 0{,}1575$
Low demand, favorable cost	$13680 - 7200$ $= 6480$	$13440 - 6400$ $= 7040$	$0{,}25 \cdot 0{,}55$ $= 0{,}1375$
Low demand, unfavorable cost	$13680 - 10080$ $= 3600$	$13440 - 8960$ $= 4480$	$0{,}25 \cdot 0{,}45$ $= 0{,}1125$

Combined probabilities are $0{,}40 \cdot 0{,}55 = 0{,}22$, $0{,}40 \cdot 0{,}45 = 0{,}18$, $0{,}35 \cdot 0{,}55 = 0{,}1925$, $0{,}35 \cdot 0{,}45 = 0{,}1575$, $0{,}25 \cdot 0{,}55 = 0{,}1375$, $0{,}25 \cdot 0{,}45 = 0{,}1125$.

State of nature	Probability	Model A	Model B
High demand, favorable cost	0.22	12600	14300
High demand, unfavorable cost	0.18	7000	9100
Medium demand, favorable cost	0.1925	9360	10560
Medium demand, unfavorable cost	0.1575	5200	6720
Low demand, favorable cost	0.1375	6480	7040
Low demand, unfavorable cost	0.1125	3600	4480

C4

$$\begin{aligned}
\text{máx}(\text{Model A}) &= \text{máx}\{12600, 7000, 9360, 5200, 6480, 3600\} = 12600,\\
\text{máx}(\text{Model B}) &= \text{máx}\{14300, 9100, 10560, 6720, 7040, 4480\} = 14300.
\end{aligned}$$

$$\text{máx}\{\text{máx}(\text{Model A}), \text{máx}(\text{Model B})\} = \text{máx}\{12600, 14300\} = 14300.$$

The Maximax choice is Model B with 14300. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\begin{aligned}
\text{mín}(\text{Model A}) &= \text{mín}\{12600, 7000, 9360, 5200, 6480, 3600\} = 3600,\\
\text{mín}(\text{Model B}) &= \text{mín}\{14300, 9100, 10560, 6720, 7040, 4480\} = 4480.
\end{aligned}$$

$$\text{máx}\{\text{mín}(\text{Model A}), \text{mín}(\text{Model B})\} = \text{máx}\{3600, 4480\} = 4480.$$

The Maximin choice is Model B with 4480 because it has the least severe worst case. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined state probabilities.

$$\begin{aligned}
EV_A &= 0{,}22(12600) + 0{,}18(7000) + 0{,}1925(9360) + 0{,}1575(5200) + 0{,}1375(6480) + 0{,}1125(3600)\\
&= 2772 + 1260 + 1801{,}2 + 819 + 891 + 405 = 7948{,}8,\\
EV_B &= 0{,}22(14300) + 0{,}18(9100) + 0{,}1925(10560) + 0{,}1575(6720) + 0{,}1375(7040) + 0{,}1125(4480)\\
&= 3146 + 1638 + 2032{,}8 + 1058{,}4 + 968 + 504 = 9347{,}2.
\end{aligned}$$

The expected value criterion chooses Model B because $9347{,}2$ is the largest expected profit. This aligns with the Maximax and Maximin outcomes for Model B, so the expected value gives the balanced decision using the probabilities.

Problem 7: Problem description

A community fitness hub must choose a membership model for the next year. Model A is a weekday focus, Model B is a family pass, and Model C is an evening pass. Demand can be high or low with probabilities $0{,}60$ and $0{,}40$. Staffing costs can be favorable or unfavorable with probabilities $0{,}55$ and $0{,}45$, and these costs are independent of demand. Expected membership counts depend on demand and model. Under high demand Model A has 700 members, Model B has 650 members, and Model C has 760 members. Under low demand Model A has 360 members, Model B has 320 members, and Model C has 420 members. Membership fees are 38 for Model A, 42 for Model B, and 34 for Model C per member per year. Variable staffing cost per member is 20 when costs are favorable and 28 when costs are unfavorable. Fixed annual costs are 8200 for Model A, 9600 for Model B, and 7000 for Model C. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Model A (weekday), Model B (family), Model C (evening)
States of Nature	Demand high or low with staffing costs favorable or unfavorable
Events	Realized demand level paired with staffing cost conditions
Consequences	Annual profit in dollars from membership fees minus staffing and fixed costs

C3

Payoff = revenue − cost, where revenue = (members × fee) and cost = (members × variable cost) + fixed cost.

Members per model by demand

Model	High	Low
Model A	700	360
Model B	650	320
Model C	760	420

Cost parameters by model

Model	Favorable	Unfavorable	Fixed cost
Model A	20	28	8200
Model B	20	28	9600
Model C	20	28	7000

Revenue parameters by member

Model	Fee per member
Model A	38
Model B	42
Model C	34

Revenue by demand

State of nature	Model A revenue	Model B revenue	Model C revenue
High demand	$700 \cdot 38 = 26600$	$650 \cdot 42 = 27300$	$760 \cdot 34 = 25840$
Low demand	$360 \cdot 38 = 13680$	$320 \cdot 42 = 13440$	$420 \cdot 34 = 14280$

Cost by demand and staffing condition

State of nature	Model A cost	Model B cost	Model C cost
High demand, favorable cost	$700 \cdot 20 + 8200 = 22200$	$650 \cdot 20 + 9600 = 22600$	$760 \cdot 20 + 7000 = 22200$
High demand, unfavorable cost	$700 \cdot 28 + 8200 = 27800$	$650 \cdot 28 + 9600 = 27800$	$760 \cdot 28 + 7000 = 28280$
Low demand, favorable cost	$360 \cdot 20 + 8200 = 15400$	$320 \cdot 20 + 9600 = 16000$	$420 \cdot 20 + 7000 = 15400$
Low demand, unfavorable cost	$360 \cdot 28 + 8200 = 18280$	$320 \cdot 28 + 9600 = 18560$	$420 \cdot 28 + 7000 = 18760$

Profit is computed as Revenue minus Cost for each alternative and state.

Profit parameters by model

State of nature	Model A	Model B	Model C	Probabilities
High demand, favorable cost	$26600 - 22200$ $= 4400$	$27300 - 22600$ $= 4700$	$25840 - 22200$ $= 3640$	$0{,}60 \cdot 0{,}55$ $= 0{,}33$
High demand, unfavorable cost	$26600 - 27800$ $= -1200$	$27300 - 27800$ $= -500$	$25840 - 28280$ $= -2440$	$0{,}60 \cdot 0{,}45$ $= 0{,}27$
Low demand, favorable cost	$13680 - 15400$ $= -1720$	$13440 - 16000$ $= -2560$	$14280 - 15400$ $= -1120$	$0{,}40 \cdot 0{,}55$ $= 0{,}22$
Low demand, unfavorable cost	$13680 - 18280$ $= -4600$	$13440 - 18560$ $= -5120$	$14280 - 18760$ $= -4480$	$0{,}40 \cdot 0{,}45$ $= 0{,}18$

Combined probabilities are $0{,}60 \cdot 0{,}55 = 0{,}33$, $0{,}60 \cdot 0{,}45 = 0{,}27$, $0{,}40 \cdot 0{,}55 = 0{,}22$, $0{,}40 \cdot 0{,}45 = 0{,}18$.

State of nature	Probability	Model A	Model B	Model C
High demand, favorable cost	0.33	4400	4700	3640
High demand, unfavorable cost	0.27	-1200	-500	-2440
Low demand, favorable cost	0.22	-1720	-2560	-1120
Low demand, unfavorable cost	0.18	-4600	-5120	-4480

C4

$$\begin{aligned}
\text{máx}(\text{Model A}) &= \text{máx}\{4400, -1200, -1720, -4600\} = 4400,\\
\text{máx}(\text{Model B}) &= \text{máx}\{4700, -500, -2560, -5120\} = 4700,\\
\text{máx}(\text{Model C}) &= \text{máx}\{3640, -2440, -1120, -4480\} = 3640.
\end{aligned}$$

$$\text{máx}\{\text{máx}(\text{Model A}), \text{máx}(\text{Model B}), \text{máx}(\text{Model C})\} = \text{máx}\{4400, 4700, 3640\} = 4700.$$

The Maximax choice is Model B with 4700. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\begin{aligned}
\text{mín}(\text{Model A}) &= \text{mín}\{4400, -1200, -1720, -4600\} = -4600,\\
\text{mín}(\text{Model B}) &= \text{mín}\{4700, -500, -2560, -5120\} = -5120,\\
\text{mín}(\text{Model C}) &= \text{mín}\{3640, -2440, -1120, -4480\} = -4480.
\end{aligned}$$

$$\text{máx}\{\text{mín}(\text{Model A}), \text{mín}(\text{Model B}), \text{mín}(\text{Model C})\} = \text{máx}\{-4600, -5120, -4480\} = -4480.$$

The Maximin choice is Model C with -4480 because it has the least severe worst case loss. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined state probabilities.

$$\begin{aligned}
EV_A &= 0{,}33(4400) + 0{,}27(-1200) + 0{,}22(-1720) + 0{,}18(-4600)\\
&= 1452 - 324 - 378{,}4 - 828 = -78{,}4,\\
EV_B &= 0{,}33(4700) + 0{,}27(-500) + 0{,}22(-2560) + 0{,}18(-5120)\\
&= 1551 - 135 - 563{,}2 - 921{,}6 = -68{,}8,\\
EV_C &= 0{,}33(3640) + 0{,}27(-2440) + 0{,}22(-1120) + 0{,}18(-4480)\\
&= 1201{,}2 - 658{,}8 - 246{,}4 - 806{,}4 = -510{,}4.
\end{aligned}$$

The expected value criterion chooses Model B because $-68{,}8$ is the largest expected profit. This matches the Maximax choice and differs from the Maximin choice, so the expected value gives a balanced decision using the probabilities.